



Your grade: 100%

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Next item →

1. Which of the following are valid steps in changing the color channel order of an image using Keras? 1 / 1 point

- ☐ Save the NumPy array directly as an image file
- ☒ Use the `array_to_image` function to convert the array back to a PIL image
- Correct

Correct! Converting the array back to a PIL image is the final step before saving.
- ☒ Change the order of the color channels in the NumPy array
- Correct

Correct! Changing the order of the color channels in the NumPy array is the key step.
- ☒ Load the image as a PIL image
- Correct

Correct! Loading the image as a PIL image is the first step.
- ☒ Convert the image to a NumPy array using `image_to_array`
- Correct

Correct! Converting the image to a NumPy array is necessary to manipulate the color channels.

2. Which function of ImageDataGenerator is commonly used to generate augmented images from a directory of images? 1 / 1 point

- ☒ `flow_from_directory`
- ☐ `fit_generator`
- ☐ `compile`
- ☐ `predict_generator`
- Correct

Correct! The function `flow_from_directory` is used to generate augmented images from a directory.

3. Which of the following are key components of a convolutional neural network (CNN)? 1 / 1 point

- ☐ Learning rate
- ☐ Optimization algorithms
- ☒ Number of feature maps
- Correct

Correct! The number of feature maps (or filters) is a key component in a CNN.
- ☒ Pooling layers
- Correct

That's right! Pooling layers are key components that help reduce spatial dimensions.
- ☒ Filter sizes
- Correct

Correct! Filter sizes are crucial in determining how features are extracted from the input.

4. Which of the following best describes the role of hyperparameters in a machine learning model? 1 / 1 point

- ☐ Hyperparameters are the data inputs to the model during training.
- ☒ Hyperparameters are used to tune the performance of a model based on the training data.
- ☐ Hyperparameters are the output predictions made by the model after training.
- ☐ Hyperparameters are the weights of the connections between neurons in a neural network.
- Correct

Correct! Hyperparameters are tuned to optimize the model's performance.

5. Which process involves adjusting the weights of a pretrained model based on new data? 1 / 1 point

- ☐ Feature selection
- ☒ Fine-tuning
- ☐ Data augmentation
- ☐ Prediction
- Correct

Correct! Fine-tuning adjusts the weights of a pretrained model to better fit the new data.